

Linear algebra: Exercise chapter 9

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1. For the following subspaces S of $\mathbb{R}^n$, find the orthogonal complement $S^\perp$:

$$S_1 = \text{span} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right), \quad S_2 = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right), \quad S_3 = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right), \quad S_4 = \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 3 \\ 2 \end{bmatrix} \right).$$

2. True or false:

- (a) The subspace spanned by $(1, 1, 0, 0, 0)$ and $(0, 0, 0, 1, 1)$ is the orthogonal complement of the subspace spanned by $(1, -1, 0, 0, 0)$ and $(2, -2, 3, 4, -4)$.
- (b) Two subspaces that only meet in the zero vector are orthogonal.
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3. Project the vector b onto the line through a :

(a) $b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$

(b) $b = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, a = \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix}.$

Compute the projection matrices for the previous exercise and use these to verify your result.

4. Project b onto the column space of A . Make use of the projection matrix.

(a) $b = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$

(b) $b = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}.$

5. Find an orthonormal basis for the space spanned by:

$$a = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad c = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}.$$

6. Consider the least square problem,

$$\min_x \frac{1}{2} \|Ax - b\|^2.$$

For the following data points:

$$\{(-1, 1), (0, 0), (1, 2), (2, 2)\},$$

- Find the closest straight line to them.
 - Compute the error.
 - Check that the error vector is perpendicular to columns of the matrix A .
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